

# Mahammad Siraj Cheruvu

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## Education

### University of South Florida, Tampa, FL

Jan 2025 – Dec 2026

Masters in Artificial Intelligence and Business Analytics | GPA: 3.88/4.0

### Shiv Nadar University, India

Aug 2020 – May 2024

Bachelor of Technology in Computer Science and Engineering | GPA: 3.16/4.0

Specialization: AI/ML | Minor: Mathematics

## Skills

- **Languages & Frameworks:** Python, Java, C#, JavaScript, R, SQL, React, Node.js, FastAPI, TensorFlow, Keras, Scikit-learn
- **Data & Databases:** PySpark, Kafka, Tableau, Power BI, Pandas, NumPy, Matplotlib, MySQL, MongoDB
- **Cloud & Tools:** AWS (Bedrock, S3, EC2), LangChain, LangGraph, Git, Docker, Postman, Visual Studio, VS Code
- **Relevant Course Work:** Machine Learning, Big Data for Business, Cloud Solution Architect, Advanced Data Base Management Systems, Probability and Statistics.
- **Certifications:** Docker Foundations Professional Certificate (2025), SnowPro Associate: Platform (2025), NLP With Sequence Models (2025), NLP With Attention Models (2025), Deep Learning - IIT Ropar (2023)

## Professional Experience

### Graduate Assistant

May 2025 - Present

University of South Florida, Tampa, FL

- Collected, cleaned, and maintained departmental data to build and sustain Tableau and Power BI analytics dashboards, enabling data-driven reporting and operational decision-making for university staff.
- Supported student onboarding and front-desk operations in a fast-paced university environment.

### Software Development Engineer, Internship

Jan 2024 – July 2024

EX Squared, Faridabad, India

- Led backend development using CLEAN architecture, building RESTful APIs with ASP.NET, reducing integration time by 30%.
- Designed and deployed an MS SQL database managing 1000+ vendor and expenses records with secure access control.
- Developed front-end features with JavaScript ES6 and collaborated to integrate vendor management, invoice generation, and document signing, improving process efficiency by 25%.
- Enabled secure file storage in Azure Blob Storage and integrated PDF generation and automated email features, cutting document turnaround time by 40%.

## Projects Experience

Aug 2023 – Dec 2025

### YouTube Automation Tool

- Built an end-to-end YouTube automation tool that takes a channel theme and topic, generates a story, creates scene and voiceover in parallel, and assembles them into a complete video for review and publishing.
- Engineered a multi-agent LangGraph pipeline with an entry agent for structured story generation and parallel agent coordination using fan-out/fan-in patterns for simultaneous media and narration synthesis.
- Integrated a human-in-the-loop review node before automated YouTube publishing, reducing manual video production effort.

### Real-Time Bus Headway Monitoring System

- Built a scalable real-time ETL pipeline using Kafka and Spark Structured Streaming to ingest, transform, and analyze high-volume event data, enabling real-time analytics and operational insights.
- Implemented a Kafka–Spark Structured Streaming pipeline with stateful event-time aggregations, deployed as a containerized multi-worker system using Spark, Kafka, MongoDB and Docker Compose for fault-tolerant, low-latency streaming.
- Enabled real-time detection of transit inefficiencies, providing real-time service alerts that can improve service reliability, operational decision-making, and passenger wait times.

### Custom AI Agent Framework & Conversational Platform

- Engineered an LLM-agnostic AI agent framework using Python with prompt orchestration for configurable roles and goals, supporting Gemini LLM and multiple LLMs with plug-and-play extensibility.
- Built and deployed a FastAPI backed with SQLite persistence, FAISS-based RAG (SentenceTransformers), and automated PDF generation, hosted on Render.
- Developed a web-based conversational platform enabling real-time agent interaction, reducing manual workflows by ~30%.